

HW6: Categorical Predictors, Interaction, and Partial F Tests

ISYE 4031 - Summer 2026 - Problems Only Version

Submission Instructions

Include your full name, Georgia Tech email, and GTID in your submitted work. You may complete the homework using JupyterLite, a local Python or R environment, spreadsheet software, handwritten work where appropriate, or another approved environment. Your submission should clearly answer each question and show enough reasoning, formulas, code output, or supporting work to be graded.

Required Data Files

- `insurance_adoption.csv`

The response is *Months*, the number of months elapsed between first adoption of an insurance innovation and adoption by the firm being considered. *Size* is total assets in millions of dollars. *Type* is *Mutual* or *Stock*; the data file also includes *Stock*, a dummy variable equal to 1 for stock companies and 0 for mutual companies.

Problem 1(a): Fit Model 1

Fit Model 1 with only size of firm as the predictor: $Months = \beta_0 + \beta_1 Size + \epsilon$.

Problem 1(b): Interpret Model 1 Coefficients

Interpret the intercept and slope in context. Note whether the intercept is practically meaningful.

Problem 1(c): Fit Model 2

Fit Model 2 with size and type of firm: $Months = \beta_0 + \beta_1 Size + \beta_2 Stock + \epsilon$.

Problem 1(d): Plot the Type Effect

Use a proper plot to show the relationship between firm type and number of months elapsed. A boxplot or side-by-side scatter/strip plot is acceptable.

Problem 1(e): Interpret Model 2 Coefficients

Interpret the coefficients in Model 2. Be explicit about the reference level for the dummy variable.

Problem 1(f): Test Type Difference and Give a 95% CI

At level 0.05, test whether the average number of months elapsed differs by firm type. Use an unadjusted two-group comparison, such as the regression model $Months \sim Stock$

or an equivalent two-sample test, and report a 95% confidence interval for the difference between Stock and Mutual firms.

This is a different question from interpreting the Stock coefficient in Model 2, because Model 2 adjusts for firm Size.

Problem 1(g): Estimate for Size = 200 and Type = Stock

Using Model 2, estimate the number of months elapsed for a stock firm with size 200. Also compute a 90% confidence interval and a 90% prediction interval.

Problem 1(h): Fit Model 3 with Interaction

Fit Model 3 with size, type, and their interaction: $Months = \beta_0 + \beta_1 Size + \beta_2 Stock + \beta_3 (Size \times Stock) + \epsilon$.

Problem 1(i): Interpret Model 3 Coefficients

Interpret the coefficients in Model 3. Explain how the size slope differs between mutual and stock firms.

Problem 1(j): Partial F Test

Run a partial F test comparing Model 1 as the reduced model and Model 2 as the complete model at level 0.05. State hypotheses, test statistic or p-value, and conclusion.

AI Usage Report

Your submission must include an AI Usage Report following the course AI policy. If you used AI tools, include the tool(s), what you asked them to help with, what output you accepted, changed, or rejected, and how you verified the final work. If prompts or screenshots materially affected your submission, include copied prompt text or screenshots in the report when appropriate. If no AI tools were used, include this statement: "No AI tools were used for this submission. You may include this report at the end of your submission or attach it as a separate file.